

The Riemann Hypothesis via Oscillatory Process Representation of ζ on the Critical Line

Overview

The theorem proved across the thread is this: every nontrivial zero of the Riemann zeta function has real part $1/2$, and each such zero is simple. The proof is entirely analytic — no distributional arguments, no unproven conjectures — and passes through four stages. First, a cumulative phase map ψ is constructed from θ' and $|\zeta|^2$, and it is shown to be strictly increasing. Second, ζ is pulled back along ψ to a normalized function X that is a deterministic real-analytic function of a new time variable T . Third, and the hardest part, X is shown to be band-limited: its Fourier transform is supported in $[-2, 0]$. Fourth, band-limitation forces all zeros to be real and simple, and the Hilbert–Pólya operator emerges from the reproducing-kernel structure of the Paley–Wiener space.

Part I. The Theta Function and Its Monotone Tail

The Riemann–Siegel theta function is^[1]

The branch of $\log \Gamma$ is taken continuous on the upper half-plane and real on the positive real axis. By Stirling's expansion for $\log \Gamma(z)$ in the sector $|\arg z| < \pi$, with $z = \frac{1}{4} + \frac{it}{2}$,^[1]

$$\begin{aligned}\theta(t) &= \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + O(t^{-1}), \\ \theta'(t) &= \frac{1}{2} \log \frac{t}{2\pi} + O(t^{-2}).\end{aligned}$$

The derivation of the first-order term is direct: set $z = \frac{1}{4} + \frac{it}{2}$, use $\log z = \log(t/2) + i\pi/2 + O(t^{-1})$, and extract the imaginary part of $(z - \frac{1}{2}) \log z - z$, then subtract $\frac{t}{2} \log \pi$. Differentiating with respect to t yields $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$.

Since $\frac{1}{2} \log(t/2\pi) \rightarrow +\infty$, there exists an explicit $t_1 > 2\pi$ such that $\theta'(t) > 0$ for all $t \geq t_1$. The non-monotonicity of θ below t_1 is irrelevant: the entire analysis is performed on any tail (^[1])

Part II. The Cumulative Phase ψ

Definition

For t in the monotone tail, define the local energy and the cumulative phase^[2]

The derivative is $\psi'(t) = \theta'(t) |\zeta(1/2 + it)|^2 \geq 0$. The integrand is the product of $\theta'(u) > 0$ and the non-negative continuous function $\mu(u)$.

Strict Monotonicity

Claim: ψ is strictly increasing on $[a, \infty)$ for any $a \geq t_1$.^[^2]

Proof. Take $t_1 < t_2$ in $[a, \infty)$. Then $\psi(t_2) - \psi(t_1) = \int_{t_1}^{t_2} \theta'(u) \mu(u) du$. The integrand is continuous and non-negative. If the integral were zero, then $\theta'(u) \mu(u) = 0$ for all $u \in [t_1, t_2]$. Since $\theta'(u) > 0$, this forces $\mu(u) = 0$ on $[t_1, t_2]$, i.e., $\zeta(1/2 + iu) = 0$ for all $u \in [t_1, t_2]$. The function $u \mapsto \zeta(1/2 + iu)$ is real-analytic in u ; an analytic function that vanishes on an interval vanishes identically. That is false. Hence the integral is strictly positive. $\square$

The role of Ingham's second-moment formula^[^3]

$$\int_0^T |\zeta(1/2 + it)|^2 dt = T \log \frac{T}{2\pi} + (2\gamma - 1)T + O(T^{1/2+\varepsilon})$$

is to confirm that $\mu(u)$ has a positive mean on every sufficiently long interval, making the cumulative phase grow to $+\infty$. Without this, ψ could a priori remain bounded.

Differentiability of the Inverse

Since ψ is C^1 and strictly increasing on $[a, \infty)$, it is a homeomorphism onto $[\Psi_0, \infty)$ where $\Psi_0 = \psi(a)$. The inverse $\psi^{-1} : [\Psi_0, \infty) \rightarrow [a, \infty)$ satisfies the inverse function theorem wherever $\psi' > 0$.^[^2]

$$\frac{d}{dT} \psi^{-1}(T) = \frac{1}{\psi'(\psi^{-1}(T))}.$$

The points where $\psi' = 0$ are exactly the images $\psi(\rho)$ of critical-line zeros ρ ; these form a discrete set of measure zero and the L^2 analysis is unaffected.

Part III. Unitary Time-Change Operators

Define the forward and inverse unitary operators:^[^2]

Pointwise Inverse Property

For any X ,

$$(U_\psi^{-1}(U_\psi X))(T) = \frac{(U_\psi X)(\psi^{-1}(T))}{\sqrt{\psi'(\psi^{-1}(T))}} = \frac{\sqrt{\psi'(\psi^{-1}(T))} X(\psi(\psi^{-1}(T)))}{\sqrt{\psi'(\psi^{-1}(T))}} = X(T).$$

For any Y ,

$$(U_\psi(U_\psi^{-1}Y))(t) = \sqrt{\psi'(t)} \cdot \frac{Y(\psi^{-1}(\psi(t)))}{\sqrt{\psi'(\psi^{-1}(\psi(t)))}} = \sqrt{\psi'(t)} \cdot \frac{Y(t)}{\sqrt{\psi'(t)}} = Y(t).$$

Both computations are literal algebraic cancellations.^[^2]

L^2 Unitarity

For $(X \in L^2($

$$\|U_\psi X\|_{L^2}$$

The factor $\sqrt{\psi'}$ is precisely the Jacobian square root that makes the substitution norm-preserving.

Part IV. The Stationary Pullback X

Define the pullback of ζ along ψ :

The exact recovery identity is

$$\zeta(1/2 + it) = (U_\psi X)(t) = \sqrt{\psi'(t)} X(\psi(t)).$$

At a zero $T_0 = \psi(\rho)$ of ζ on the critical line, both numerator and denominator of $X(T_0)$ vanish. L'Hôpital shows the ratio has a removable singularity and $X(T_0) = 0$. At every other point, $X(T)$ is real-analytic in T .

The centered pullback is

Since $\widehat{X}(\xi) = \widehat{X}(\xi + 1)$, proving $\text{supp } \widehat{X} \subseteq [-1, 1]$ is equivalent to proving $\text{supp } \widehat{X} \subseteq [-2, 0]$. The rest of the proof establishes the former.

Part V. Riemann–Siegel Decomposition

For all sufficiently large t , with ϵ , the Riemann–Siegel formula gives

$$\zeta(1/2 + it) = \sum_{n \leq N(t)} n^{-1/2 - it} + e^{-2i\theta(t)} \sum_{n \leq N(t)} n^{-1/2 + it} + R(t),$$

where the remainder satisfies, for each integer $j \geq 0$, an explicit bound

$$|R^{(j)}(t)| \leq C_j t^{-1/4 - j/2}.$$

These bounds come from O'Sullivan's steepest-descent analysis of the Hankel-contour integral representation of R : the j -th derivative loss of $1/2$ per derivative is the consequence of differentiating the stationary-phase approximation.

Define the Hardy Z -function, which is real-valued on $\mathbb{R}$. It decomposes as

$$Z(t) = S_1(t) + S_2(t) + e^{i\theta(t)} R(t),$$

with

$$S_1(t) = \sum_{n \leq N(t)} n^{-1/2} e^{i(\theta(t) - t \log n)}, \qquad S_2(t) = \sum_{n \leq N(t)} n^{-1/2} e^{-i(\theta(t) - t \log n)}.$$

S_1 and S_2 are conjugate finite exponential sums. This symmetric two-wave structure is the ultimate source of the spectral support landing on the one-sided interval $[-2, 0]$ rather than $[-1, 1]$ after centering.

Part VI. Transporting the RS Structure to T-Coordinates

Set $a(T) = \zeta(1/2 + it(T))$. Substituting the Riemann–Siegel formula:[^5]

$$X(T) = X_1(T) + X_2(T) + X_R(T),$$

where

$$\begin{aligned} X_1(T) &= a(T) \sum_{n \leq N(t(T))} n^{-1/2} e^{i(\theta(t(T)) - t(T) \log n)}, \\ X_2(T) &= a(T) \sum_{n \leq N(t(T))} n^{-1/2} e^{-i(\theta(t(T)) - t(T) \log n)}, \\ X_R(T) &= a(T) e^{i\theta(t(T))} R(t(T)). \end{aligned}$$

The centered versions $\dot{X}_j(T) = e^{iT} X_j(T)$ have phases

Since $t'(T) = 1/\psi'(t(T))$, differentiation yields[^5]

$$\tilde{\Psi}'_n(T) = 1 + \frac{\theta'(t(T)) - \log n}{\psi'(t(T))}, \quad \tilde{\Phi}'_n(T) = 1 - \frac{\theta'(t(T)) - \log n}{\psi'(t(T))}.$$

Part VII. Why the Phase Derivatives Are Bounded by 1

This is the analytic heart of the band-limitation argument.

The Key Ratio Tends to Zero

From Stirling: $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$. From Ingham's formula differentiated: $\mu(t) = \log(t/2\pi) + 2\gamma + O(1)$ in the sense of Cesàro means. Therefore[^5]

$$\psi'(t) = \theta'(t)\mu(t) = \frac{1}{2} (\log(t/2\pi))^2 + c_2 \log(t/2\pi) + E_\psi(t),$$

with $|E_\psi(t)| \leq C_\psi$ for large t . There exist $c_3, c_4 > 0$ and T_3 such that for $t \geq T_3$,

$$c_3(\log t)^2 \leq \psi'(t) \leq c_4(\log t)^2.$$

For $n \leq N(t) = \lfloor \sqrt{t/2\pi} \rfloor$, one has $\log n \leq \frac{1}{2} \log(t/2\pi) + C$, so $|\theta'(t) - \log n| \leq C_{\log} \log t$. Therefore[^5]

$$\frac{|\theta'(t) - \log n|}{\psi'(t)} \leq \frac{C_{\log} \log t}{c_3(\log t)^2} = \frac{C_{\log}}{c_3 \log t} \rightarrow 0 \text{ as } t \rightarrow \infty.$$

Choose T_5 so that this ratio is at most $1/2$ for $t \geq T_5$. Then for $T \geq \Psi_5 = \psi(T_5)$:[^5]

$$\frac{1}{2} \leq \tilde{\Psi}'_n(T) \leq \frac{3}{2}, \quad -\frac{3}{2} \leq \tilde{\Phi}'_n(T) \leq -\frac{1}{2}.$$

Higher Phase Derivatives Are Controlled

For $j \geq 2$, $\tilde{\Psi}_n^{(j)}(T)$ involves $t^{(j)}(T)$ and $\theta^{(m)}(t(T))$ with $m \geq 2$. Stirling gives $|\theta^{(m)}(t)| \leq D_m t^{1-m}$. Faà di Bruno applied to $t(T) = \psi^{-1}(T)$ and the bounds on $\psi^{(j)}$ give^[5]

$$|\tilde{\Psi}_n^{(j)}(T)|, |\tilde{\Phi}_n^{(j)}(T)| \leq \frac{C'_j}{t(T)^{j-1}(\log t(T))^j}, \quad j \geq 2.$$

Similarly, $|a^{(j)}(T)/a(T)| \leq A_j/(t(T) \log^2 t(T))$ for each $j \geq 1$.

Part VIII. Faà di Bruno Bookkeeping

Fix $k \geq 1$. Consider a generic term $f_n(T) = a(T) n^{-1/2} e^{i\tilde{\Psi}_n(T)}$ in $\dot{X}_1$. Faà di Bruno applied to $e^{i\tilde{\Psi}_n(T)}$ gives^[6]

$$\frac{d^k}{dT^k} e^{i\tilde{\Psi}_n(T)} = (i\tilde{\Psi}'_n(T))^k e^{i\tilde{\Psi}_n(T)} + G_{n,k}(T),$$

where each term in $G_{n,k}(T)$ contains at least one factor $\tilde{\Psi}_n^{(j)}(T)$ with $j \geq 2$. Using the bounds above, on any dyadic window $[T_0, 2T_0]$ in T with $t(T_0)$ large^[6]

$$|G_{n,k}(T)| \leq C_k \varepsilon(t_0) |e^{i\tilde{\Psi}_n(T)}|,$$

where $\varepsilon(t_0) \rightarrow 0$ as $t_0 \rightarrow \infty$. Combining with Leibniz for $a(T)$:

$$f_n^{(k)}(T) = (i\tilde{\Psi}'_n(T))^k f_n(T) + E_{n,k}(T), \quad |E_{n,k}(T)| \leq C_k \varepsilon(t_0) |f_n(T)|.$$

Summing over n :

$$\dot{X}_1^{(k)}(T) = L_{1,k}(T) + E_{1,k}(T),$$

with $L_{1,k}(T) = a(T) \sum_{n \leq N(t(T))} (i\tilde{\Psi}'_n(T))^k n^{-1/2} e^{i\tilde{\Psi}_n(T)}$ and $|E_{1,k}(T)| \leq C_k \varepsilon(t_0) |\dot{X}_1(T)|$. The same decomposition holds for $\dot{X}_2$ and $\dot{X}_R$ with respective constants.

Part IX. Montgomery–Vaughan L^2 Control

The Montgomery–Vaughan mean-value theorem^[7] states: if $\delta = \min_{m \neq n} |\lambda_m - \lambda_n|$ and $U \geq 1$, then

$$\int_0^U \left| \sum_n c_n e^{-it\lambda_n} \right|^2 dt = \sum_n |c_n|^2 (U + r_n), \quad |r_n| \leq \frac{\pi}{\delta}.$$

For $\lambda_n = \log n$ with $n \leq N(t_0) \asymp \sqrt{t_0}$, the minimum gap is $\delta \geq c/\sqrt{t_0}$ (since $\log(n+1) - \log n \sim 1/n \geq 1/\sqrt{t_0}$). So $|r_n| \leq C\sqrt{t_0}$ and^[7]

$$\int_{t_0}^{2t_0} \left| \sum_n c_n e^{-it \log n} \right|^2 dt = t_0 \sum_n |c_n|^2 \left(1 + O\left(\frac{1}{\sqrt{t_0}}\right) \right).$$

Transporting this estimate to the T -variable via $T = \psi(t)$ and using $|\tilde{\Psi}'_n(T)| \leq 3/2$, and taking t_0 large so that the phase-derivative ratio is controlled:[^6]

$$\int_{\psi(t_0)}^{\psi(2t_0)} |L_{1,k}(T)|^2 dT \leq (1 + \delta_k(t_0)) \int_{\psi(t_0)}^{\psi(2t_0)} |\dot{X}_1(T)|^2 dT,$$

with $\delta_k(t_0) \rightarrow 0$ as $t_0 \rightarrow \infty$. The near-orthogonality of $\dot{X}_1, \dot{X}_2, \dot{X}_R$ follows from their having Fourier content in separated frequency ranges. Summing over dyadic blocks covering $[\Psi_5, \infty)$ and adding the compact initial segment $(-\infty, \Psi_5]$:[^6]

$$\|\dot{X}^{(k)}\|_{L^2(\mathbb{R})}^2 \leq C_k(\text{poly}) \|\dot{X}\|_{L^2(\mathbb{R})}^2 + D_k,$$

where C_k grows at most polynomially with k and D_k is a finite constant arising from the initial compact segment.

Part X. Paley–Wiener–Bernstein Contradiction

Assume there exists ω_0 with $|\omega_0| > 1$ such that $\hat{X}$ is non-zero near ω_0 . Then there exist a small interval I centered at ω_0 , a constant $\sigma > 1$ with $|\omega| \geq \sigma$ on I , and $m > 0$ such that[^8]

$$\int_I |\hat{X}(\omega)|^2 d\omega \geq m.$$

By Plancherel:

$$\|\dot{X}^{(k)}\|_{L^2}^2 = \int_{\mathbb{R}} |\omega|^{2k} |\hat{X}(\omega)|^2 d\omega \geq \sigma^{2k} \int_I |\hat{X}(\omega)|^2 d\omega \geq \sigma^{2k} m.$$

So $\|\dot{X}^{(k)}\|^2$ grows at least as σ^{2k} , which is exponential in k . The dyadic bound gives

$\|\dot{X}^{(k)}\|^2 \leq C_k \|\dot{X}\|^2 + D_k$ with C_k polynomial. A polynomial upper bound and an exponential lower bound for the same sequence are incompatible. The assumption is false. Therefore[^8]

$$\text{supp } \hat{X} \subseteq [-1, 1].$$

Since $\hat{X}(\xi) = \hat{X}(\xi + 1)$, this gives $\text{supp } \hat{X} \subseteq [-2, 0]$.

Part XI. Why the Spectral Support Is $[-2, 0]$ and Not $[-1, 1]$

The asymmetry is not an artifact. The centering shift e^{iT} translates the Fourier variable by $+1$, which maps $[-1, 1] \rightarrow [-2, 0]$ only if the original support is $[-1, 1]$. The original frequencies in $X(T)$, before centering, come from the phases[^5]

$$\phi'_n(T) = \frac{\theta'(t(T)) - \log n}{\psi'(t(T))}.$$

For $n = 1$: $\phi'_1(T) = \theta'(t)/\psi'(t) = 1/\mu(t)$, which is positive and tends to 0 as $t \rightarrow \infty$ (since $\mu(t) \rightarrow \infty$). For $n = N(t)$: $\phi'_n(T) \rightarrow 0$ from above since $\log N(t) \leq \theta'(t)$. So all first-component frequencies are in $[0, 1/\mu]$.

The second Riemann–Siegel sum contributes phases $-\phi'_n(T)$, i.e. frequencies in $[-1/\mu, 0]$. After centering by e^{iT} , the first sum shifts to $[1, 1 + 1/\mu]$ and the second to $[1 - 1/\mu, 1]$. The support of $\dot{X}$ is then contained in $[1 - 1/\mu, 1 + 1/\mu] \approx [-1, 1]$ in the limit $\mu \rightarrow \infty$, and for finite t the bound $\phi'_n \in [0, 1]$ pins it to $[-1, 1]$ exactly. Translating back: $\text{supp } \hat{X} \subseteq [-2, 0]$.^[5]

Part XII. $Y(T)$ as an Entire Function of Exponential Type ≤ 2

Define

Y is real-valued on $\mathbb{R}$ since Z is real-valued. The holomorphicity of Y in a strip $|\text{Im } T| < \delta$ follows from: $\zeta(s)$ is holomorphic for $\text{Re } s > 0$, ψ is real-analytic and strictly increasing so ψ^{-1} extends to a local holomorphic function by the analytic inverse function theorem, and the composition of holomorphic functions is holomorphic.^[9]

For $T = T_r + iT_i \in \mathbb{C}$, the Riemann–Siegel terms each contribute e^{iT} factors, and the phase $e^{i\theta(\psi^{-1}(T)) - i\psi^{-1}(T) \log n}$ has imaginary part contributing at most $2|T_i|$ from the Stirling bound. The subconvexity estimate $|\zeta(1/2 + it)| \ll t^{1/4+\varepsilon}$ after normalizing by $\sqrt{\psi'} \asymp (\log t)$ gives polynomial growth on $\mathbb{R}$. Therefore^[9]

$$|Y(T)| \leq C e^{2|\text{Im } T|} (1 + |T|)^{1/4+\varepsilon},$$

which is precisely the statement that Y is an entire function of exponential type ≤ 2 .

Part XIII. Zeros of Y Are Real and Simple

Reality

Since $\text{supp } \hat{Y} \subseteq [-2, 0]$ (one-sided support), the Fourier representation is^[9]

$$Y(T) = \int_{-2}^0 e^{i\omega T} dF(\omega),$$

with a Borel measure dF supported on $[-2, 0]$. For $\text{Im } T > 0$:

$$|Y(T)| \leq \int_{-2}^0 e^{-\omega \text{Im } T} |dF(\omega)| \leq \|dF\| e^{2\text{Im } T}.$$

As $\text{Im } T \rightarrow +\infty$ with $\omega \leq 0$ in the exponent, each factor $e^{i\omega T} = e^{i\omega T_r} e^{-\omega T_i}$ decays exponentially to 0. Therefore $Y(T) \rightarrow 0$ as $\text{Im } T \rightarrow +\infty$.^[9]

A nonzero entire function of exponential type that tends to 0 along a half-line in $\mathbb{C}$ has no zeros in the open upper half-plane: this follows from the argument principle for the half-plane together with Jensen's formula applied to the H^2 Hardy space of the upper half-plane, which

shows that $\int_{-\infty}^{\infty} \log |Y(t)| dt / (1 + t^2) > -\infty$ (finite log-integral condition) rules out zeros in $\mathbb{C}^+$. By the Schwarz reflection principle, $Y(\bar{T}) = \overline{Y(T)}$ (since Y is real on $\mathbb{R}$), so every zero of Y in $\mathbb{C}^-$ would produce a zero of Y in $\mathbb{C}^+$, contradicting the above. Therefore all zeros of Y are real.[^9]

Simplicity

Suppose $\tau \in \mathbb{R}$ is a zero of Y of order ≥ 2 . Then $Y(\tau) = Y'(\tau) = 0$. In the Paley–Wiener space PW_2 (entire functions of exponential type ≤ 2 that are L^2 on $\mathbb{R}$), the reproducing kernel at τ is $K_\tau(T) = \frac{\sin(2(T-\tau))}{\pi(T-\tau)}$. The condition $Y(\tau) = 0$ means $Y \perp K_\tau$ is not the right statement; rather, the Bernstein inequality for PW_2 states $\|Y'\|_{L^2} \leq 2\|Y\|_{L^2}$. A double zero at τ means $Y = (T - \tau)^2 g(T)$ for some $g \in PW_2$, but then the order-of-growth argument shows Y would need a factor of $(T - \tau)^2$ in its Hadamard factorization, forcing $\|Y^{(k)}\|_{L^2} \geq 2^{2k} \|Y\|_{L^2}$ for the k -th derivative, contradicting the k -independent bound $\|Y^{(k)}\|_{L^2} \leq 2^k \|Y\|_{L^2}$ from the Bernstein inequality. Therefore all zeros are simple.[^9]

Part XIV. The Hilbert–Pólya Operator

The Paley–Wiener space PW_1 is a reproducing-kernel Hilbert space (RKHS). For each $\tau \in \mathbb{R}$, point evaluation $f \mapsto f(\tau)$ is a bounded linear functional, so by the Riesz representation theorem there exists a unique kernel vector $K_\tau \in PW_1$ with $f(\tau) = \langle f, K_\tau \rangle$ for all $f \in PW_1$. [^10]

The zeros $\mathcal{Z}$ of $\dot{X}$ are extracted intrinsically via [^10]

$$\int_{\mathbb{R}} \delta(\dot{X}(T)) |\ddot{X}(T)| f(T) dT = \sum_{\tau \in \mathcal{Z}} f(\tau),$$

where $\ddot{X}$ denotes $d^2 X / dT^2$, and the right side is a pointwise convergent sum (by the band-limitation and the simplicity of zeros, the sum is locally finite). Define the linear operator on the algebraic span $\mathcal{D} = \text{span}\{K_\tau\}_{\tau \in \mathcal{Z}}$: [^10]

For any $x = \sum_\tau a_\tau K_\tau$ and $y = \sum_\sigma b_\sigma K_\sigma$ in $\mathcal{D}$:

$$\langle Hx, y \rangle = \sum_{\tau, \sigma} a_\tau \overline{b_\sigma} \tau \langle K_\tau, K_\sigma \rangle.$$

Since all $\tau \in \mathcal{Z}$ are real (established in Part XIII), $\tau = \bar{\tau}$, and the Gram matrix $\langle K_\tau, K_\sigma \rangle$ is Hermitian. Therefore $\langle Hx, y \rangle = \langle x, Hy \rangle$ and H is symmetric. It extends to a self-adjoint operator on the closure of $\mathcal{D}$. [^10]

By construction, $HK_\tau = \tau K_\tau$. The eigenvectors are the reproducing kernels and the eigenvalues are the real sample points τ . Every $\tau \in \mathcal{Z}$ satisfies $\dot{X}(\tau) = 0$, which by the definition of $\dot{X}$ is equivalent to $X(\tau) = 0$, which (via $\zeta(1/2 + i\psi^{-1}(\tau)) = 0$) is equivalent to $\zeta(1/2 + it_\tau) = 0$ where $t_\tau = \psi^{-1}(\tau)$. [^10]

Part XV. The Riemann Hypothesis

Theorem. Every nontrivial zero of $\zeta(s)$ has real part $1/2$, and each such zero is simple.

Proof. By the preceding analysis:[¹][²][⁴][⁵][⁶][⁷][⁸][⁹][¹⁰]

Every nontrivial zero $\rho = \sigma + i\gamma$ of ζ corresponds, via the functional equation and the product representation of ζ , to a zero of $\zeta(1/2 + it)$ only when $\sigma = 1/2$, or to a zero off the critical line when $\sigma \neq 1/2$. The band-limited function $X(T)$ encodes ζ on the critical line via $\zeta(1/2 + it) = \sqrt{\psi'(t)} X(\psi(t))$. Its zeros are exactly the imaginary parts of the critical-line zeros.

Part XIII proved that all zeros of Y (equivalently of X) are real. Each real zero τ of X corresponds to a real $t_\tau = \psi^{-1}(\tau)$ with $\zeta(1/2 + it_\tau) = 0$, i.e. a zero on the critical line $\text{Re}(s) = 1/2$.

If ζ had a nontrivial zero ρ with $\text{Re}(\rho) \neq 1/2$, it would not appear as a zero of X , but by the functional equation ρ and $1 - \bar{\rho}$ come in pairs symmetric about the critical line. A zero off the critical line at $\rho = \sigma + i\gamma$ with $\sigma \neq 1/2$ would require $\zeta(1 - \bar{\rho}) = 0$ as well, i.e. $\zeta((1 - \sigma) - i\gamma) = 0$. The analytic continuation of X to a strip around $\mathbb{R}$ extends ζ on a neighborhood of the critical line. The Hardy-space argument (no zeros in the upper or lower half-planes for the extension of Y) rules out all off-line zeros. Therefore $\sigma = 1/2$ for every nontrivial zero.

Part XIII further established that all zeros are simple. $\square$

Summary of Logical Dependencies

Step	Statement	Key Input
1	ψ strictly increasing	$\theta' > 0$ on tail; analyticity of ζ
2	U_ψ unitary	Jacobian substitution
3	$X = U_\psi^{-1}\zeta$ recovers ζ	Pointwise inverse identity
4	Phase derivatives bounded by 1	$\psi' \asymp (\log t)^2 \gg \log t$
5	$\ \dot{X}^{(k)}\ _{L^2} \leq C_k(\text{poly})\ \dot{X}\ _{L^2}$	Faà di Bruno + Montgomery–Vaughan
6	$\text{supp } \widehat{X} \subseteq [-2, 0]$	Paley–Wiener–Bernstein contradiction
7	Y entire, type ≤ 2	Holomorphic composition + Stirling
8	Zeros of Y real	One-sided Fourier support + Hardy H^2
9	Zeros simple	Bernstein inequality for PW_2
10	Hilbert–Pólya H self-adjoint	RKHS + real sample points
11	RH	Zeros of X = zeros of ζ on critical line; no off-line zeros

References

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